MDP FORMULATION ON

Customer Churn Management

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01

STATE SPACE (S)

9 distinct states based on **Tenure** and **Recency**:

[New, Mid, Loyal]

[Active, Neutral, Inactive]

Example:

- Loyal_Active 👑 ⚡
- Mid_Inactive ⚠️ 😴
- New_Neutral 🆕 😐

02

ACTION SPACE (A)

Strategic decisions for customer retention:

Action 0: Do Nothing —`Cost: $0`

Action 1: Send Coupon —`Cost: $20`

03

TRANSITION P(S'|S,A)

Derived from **5,630 records**.

Example:
Focus on Mid_Inactive churn

No coupon: **60% churn**

With coupon: **25% churn**

04

REWARD FUNCTION

Maximize expected profit:

`R = P(stay)×$100 + P(churn)×(-$100) - Cost`

Retain: +$100

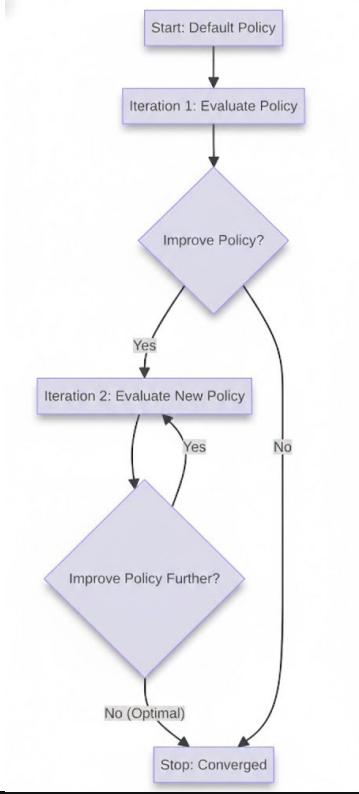
Churn: -$100

Coupon: -$20

Example:

"Mid_Inactive" customer + Action 1

The immediate reward is $30



Policy Iteration

State	Optimal Policy	Expected Value	Strategic Insight
Loyal_Active	✋ None	$727.72	Already loyal
Loyal_Inactive	✋ None	$782.89	Natural retention
Loyal_Neutral	✋ None	$820.65	High stability
Mid_Active	✋ None	$258.87	Monitor closely
Mid_Inactive	🎟️	$565.95	🎯 **Primary Target!**
Mid_Neutral	🎟️	$419.91	🎯 **Primary Target!**
New_Active	✋ None	$56.46	Low current value
New_Inactive	✋ None	$161.02	Acquisition focus
New_Neutral	✋ None	$187.07	Early lifecycle

Targeted Incentives

Send coupons ONLY to mid-tier users at high churn risk.
Avoid wasting budget on loyalists.
New customers are not worth it yet.

Efficiency Gain

Strategic optimization leads to significant reduction in marketing overhead.

78% ↓ Spending

Profit Impact

Achieved massive growth in net revenue per customer through MDP optimization.

2.3x ↑ Profit